Física II – Aula 18

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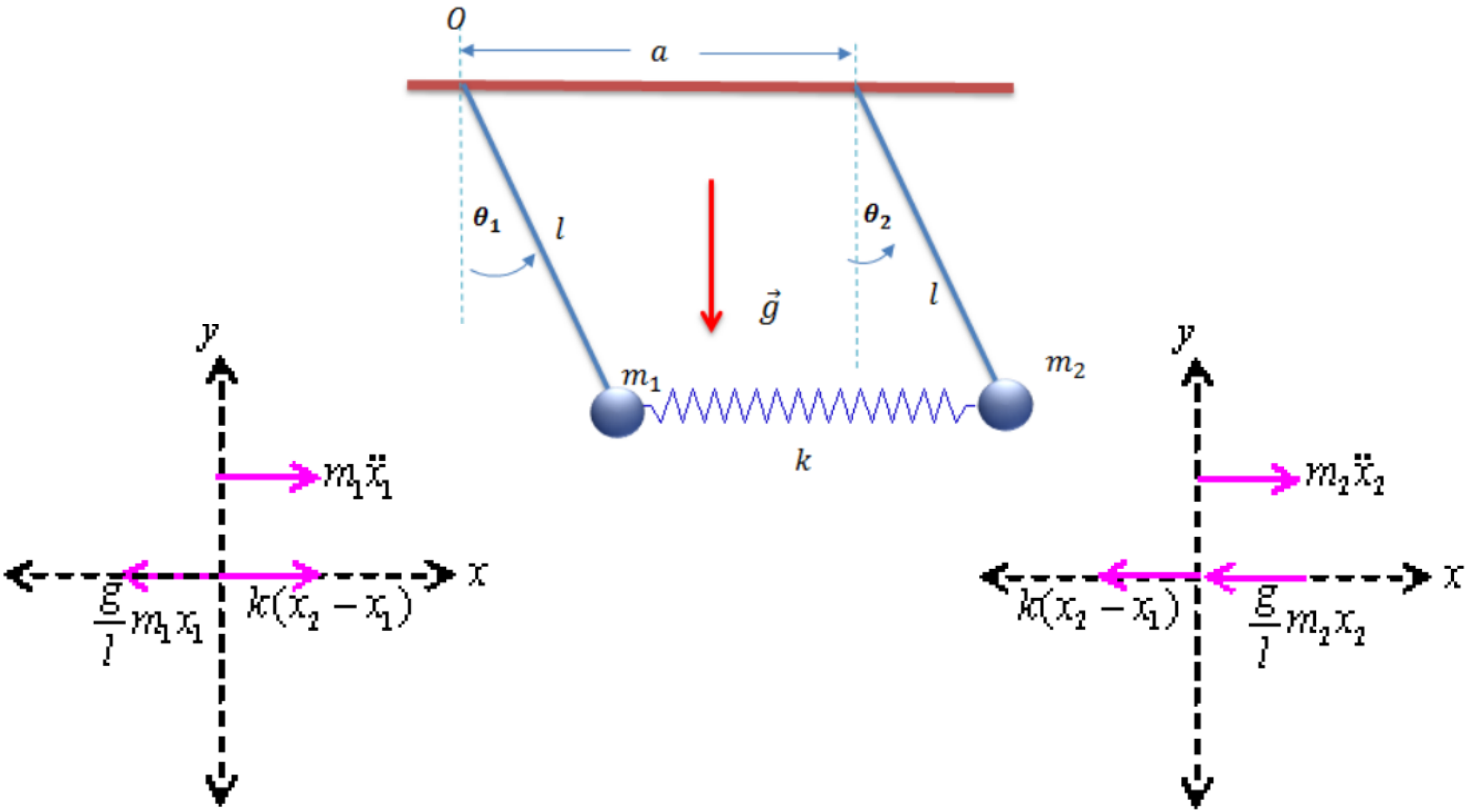
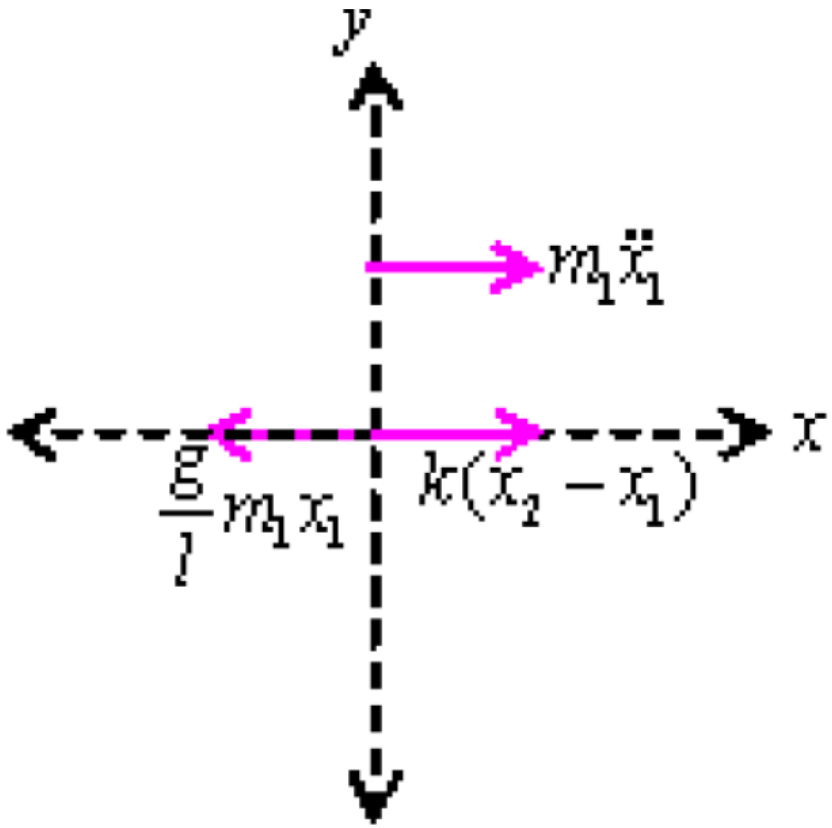
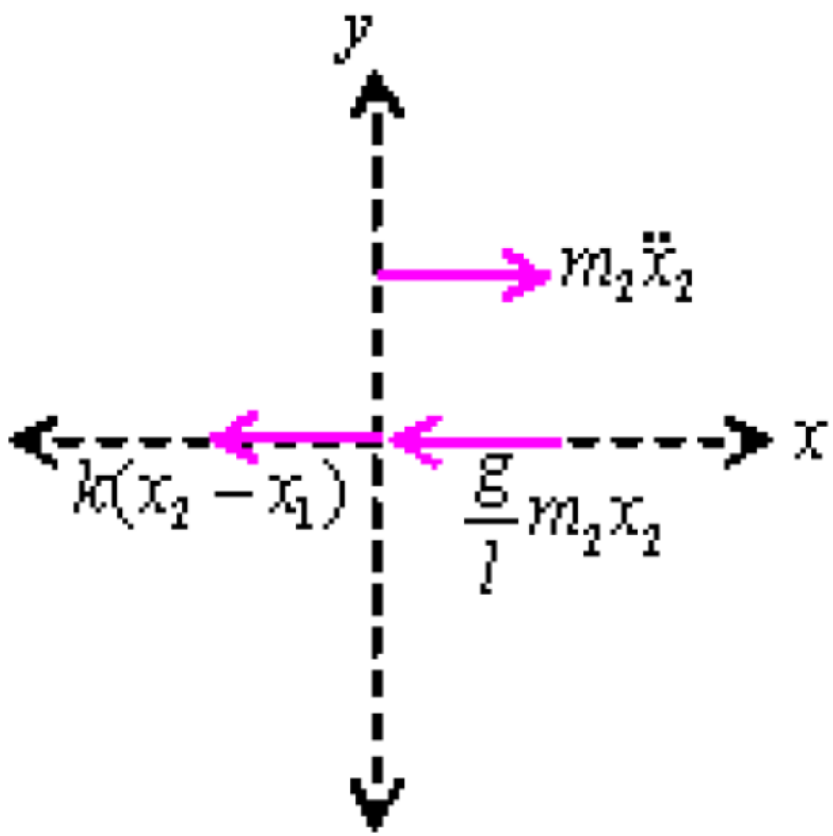

O
a
θ1
l
θ2
l
g
m1
m2
k
y
m1ẍ1
x
g/l m1x1
k(x2 − x1)
y
m2ẍ2
x
k(x2 − x1)
g/l m2x2



$$-\frac{g}{l}m_1x_1+k\left(x_2-x_1\right)=m_1\ddot{x}_1 \qquad (1)$$

$$m_1\ddot{x}_1+\frac{g}{l}m_1x_1-k\left(x_2-x_1\right)=0 \qquad (2)$$

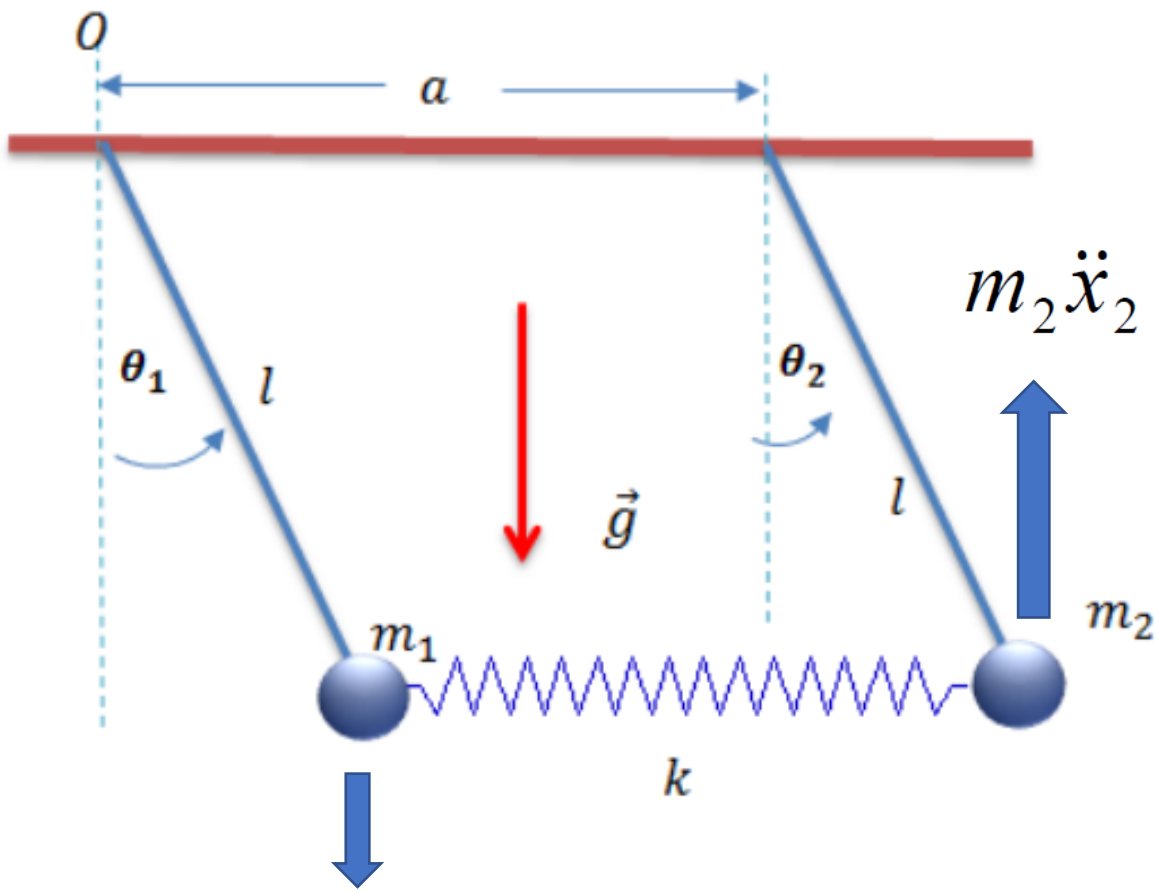
$$m_1\ddot{x}_1+\left(\frac{g}{l}m_1+k\right)x_1-kx_2=0 \qquad (3)$$



$$-\frac{g}{l}m_2x_2 - k(x_2 - x_1) = m_2\ddot{x}_2 \qquad (4)$$

$$m_2\ddot{x}_2 + \frac{g}{l}m_2x_2 + k(x_2 - x_1) = 0 \qquad (5)$$

$$m_2\ddot{x}_2 + \left(\frac{g}{l}m_2 + k\right)x_2 - kx_1 = 0 \qquad (6)$$



$$m_2\ddot{x}_2 + \left(\frac{g}{l}m_2 + k\right)x_2 - kx_1 = 0 \qquad (6)$$

$$m_1\ddot{x}_1 + \left(\frac{g}{l}m_1 + k\right)x_1 - kx_2 = 0 \qquad (3)$$

$$\ddot{x}_1 + \frac{g}{l}x_1 - \frac{k}{m_1}(x_2 - x_1) = 0 \qquad (7)$$

$$\ddot{x}_2 + \frac{g}{l}x_2 + \frac{k}{m_2}(x_2 - x_1) = 0 \qquad (8)$$

$m_1 = m_2 = m$,

$$\ddot{x}_1 + \frac{g}{l}x_1 - \frac{k}{m}(x_2 - x_1) = 0 \qquad (9)$$

$$\ddot{x}_2 + \frac{g}{l}x_2 + \frac{k}{m}(x_2 - x_1) = 0 \qquad (10)$$

$$\ddot{x}_1 + \frac{g}{l}x_1 - \frac{k}{m}(x_2 - x_1) = 0 \qquad (9)$$

$$+$$

$$\ddot{x}_2 + \frac{g}{l}x_2 + \frac{k}{m}(x_2 - x_1) = 0 \qquad (10)$$

$$(\ddot{x}_1 + \ddot{x}_2) + \frac{g}{l}(x_1 + x_2) = 0 \qquad (11)$$

$$X_1 = x_1 + x_2$$

$$\ddot{X}_1 = \ddot{x}_1 + \ddot{x}_2$$

$$\longrightarrow \quad \ddot{X}_1 + \frac{g}{l}X_1 = 0 \qquad (12)$$

$$\ddot{X}_1 + \frac{g}{l} X_1 = 0 \qquad (12)$$

$$\omega_1^2 = \frac{g}{l} \qquad (13)$$

$$\boxed{X_1(t) = A_1 \sin(\omega_1 t + \varphi_1)} \qquad (14)$$

$$\ddot{x}_1 + \frac{g}{l}x_1 - \frac{k}{m}(x_2 - x_1) = 0 \qquad (9)$$

$$-\quad \ddot{x}_2 + \frac{g}{l}x_2 + \frac{k}{m}(x_2 - x_1) = 0 \qquad (10)$$

$$(\ddot{x}_1 - \ddot{x}_2) + \frac{g}{l}(x_1 - x_2) - \frac{2k}{m}(x_2 - x_1) = 0$$

$$X_2 = x_1 - x_2$$

$$\ddot{X}_2 = \ddot{x}_1 - \ddot{x}_2$$

$$\ddot{X}_2 + \frac{g}{l}X_2 + \frac{2k}{m}X_2 = 0 \qquad (16)$$

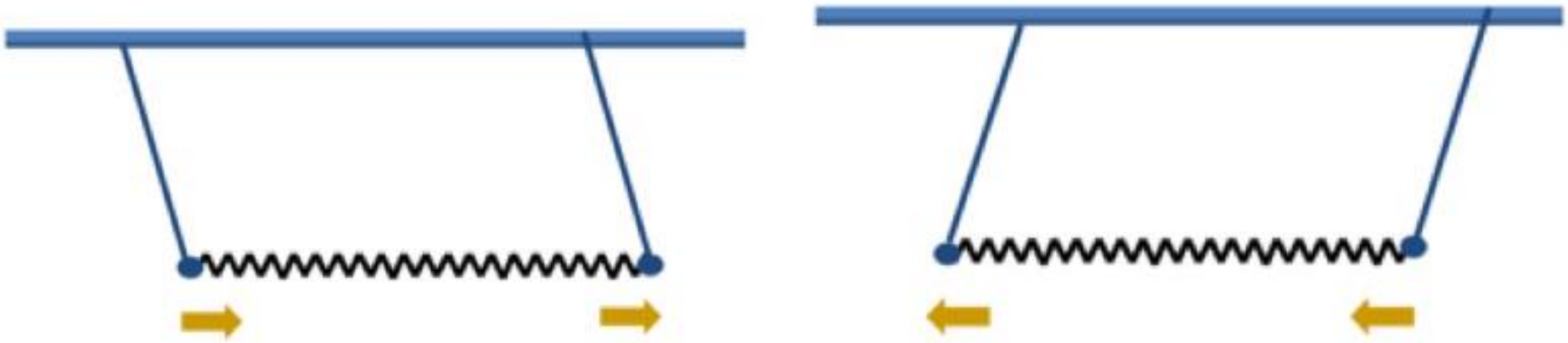
$$\ddot{X}_2 + \left(\frac{g}{l} + \frac{2k}{m}\right)X_2 = 0 \qquad (17)$$

$$\ddot{X}_2 + \left(\frac{g}{l} + \frac{2k}{m}\right) X_2 = 0 \qquad (17)$$

$$\omega_2^2 = \frac{g}{l} + \frac{2k}{m}$$

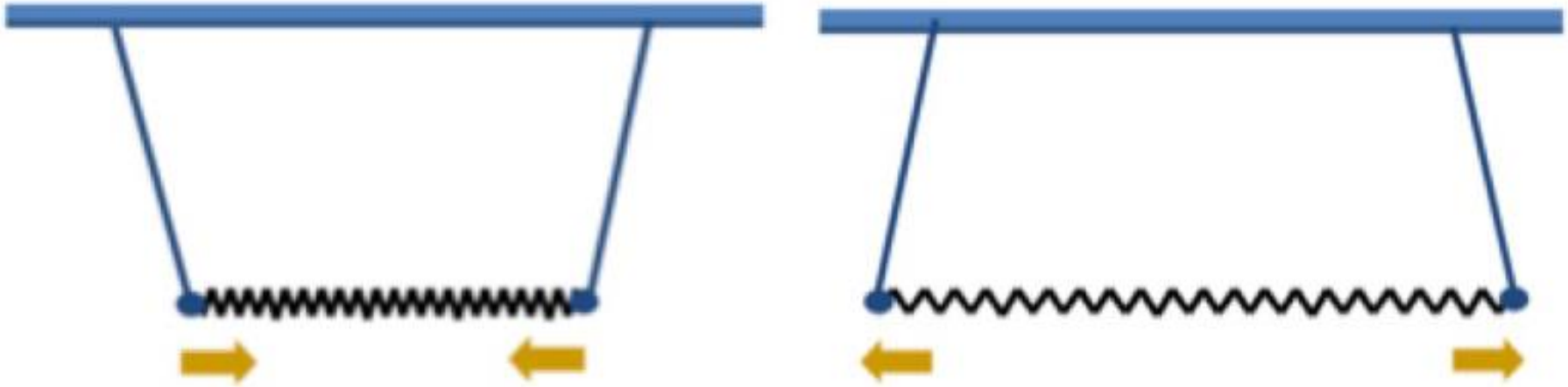
$$\boxed{X_2(t) = A_2 \sin(\omega_2 t + \varphi_2) \qquad (18)}$$

Modo 1

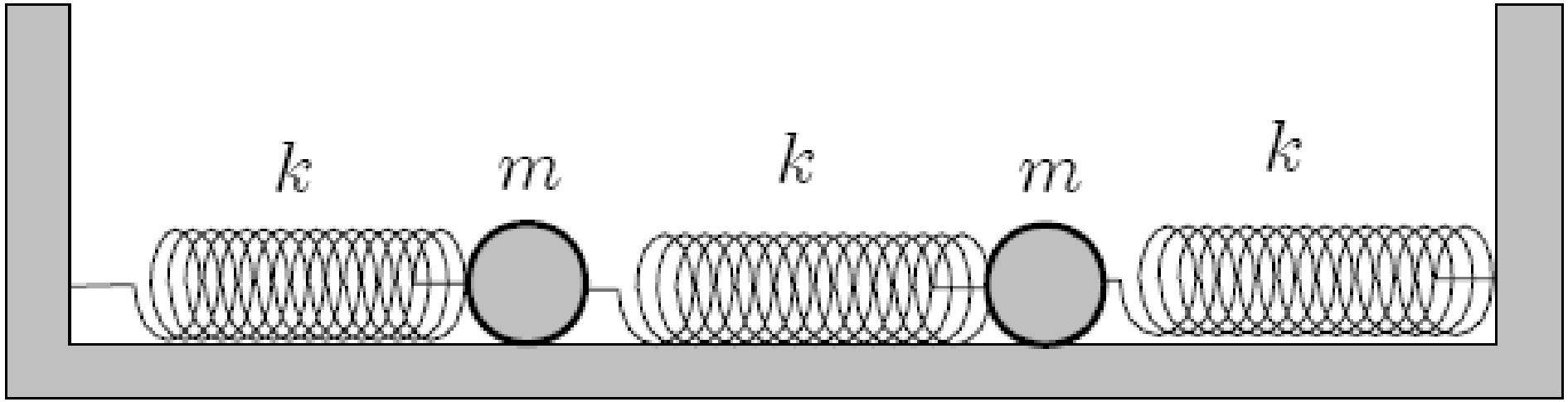


$$\omega_1 = \sqrt{\frac{g}{l}}.$$

Modo 2



$$\omega_2 = \sqrt{\frac{g}{l} + \frac{2k}{m}}.$$



$$
\begin{aligned}
m\ddot{x}_1 &= k(x_2 - x_1) - kx_1, \\
m\ddot{x}_2 &= -k(x_2 - x_1) - kx_2,
\end{aligned}
$$

$\omega^2 = k/m$

$$
\begin{aligned}
\ddot{x}_1 &= \omega^2(x_2 - 2x_1), \\
\ddot{x}_2 &= -\omega^2(2x_2 - x_1)
\end{aligned}
$$

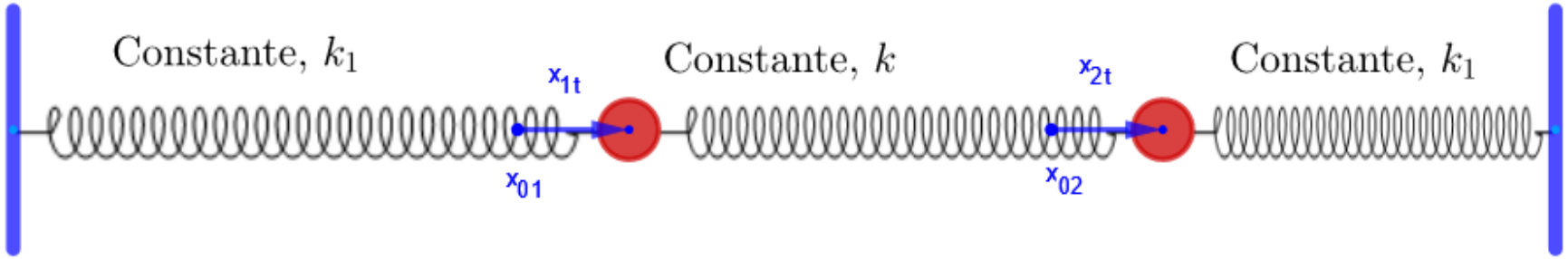
$$\begin{cases} \ddot{x}_1 &= \omega^2(x_2 - 2x_1), \\ \ddot{x}_2 &= -\omega^2(2x_2 - x_1) \end{cases}$$

$$\begin{cases} \ddot{x}_2 - \ddot{x}_1 &= -3\omega^2\left(x_2 - x_1\right) \\ \ddot{x}_2 + \ddot{x}_1 &= -\omega^2\left(x_2 + x_1\right), \end{cases}$$

$$\omega_1 = \sqrt{\frac{k}{m}}, \qquad \omega_2 = \sqrt{3\frac{k}{m}}.$$

https://www.geogebra.org/m/uv2rapqq

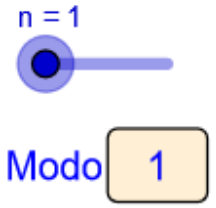
$$\begin{cases} \varsigma_1 = x_2 + x_1 \\ \varsigma_2 = x_2 - x_1 \end{cases}$$

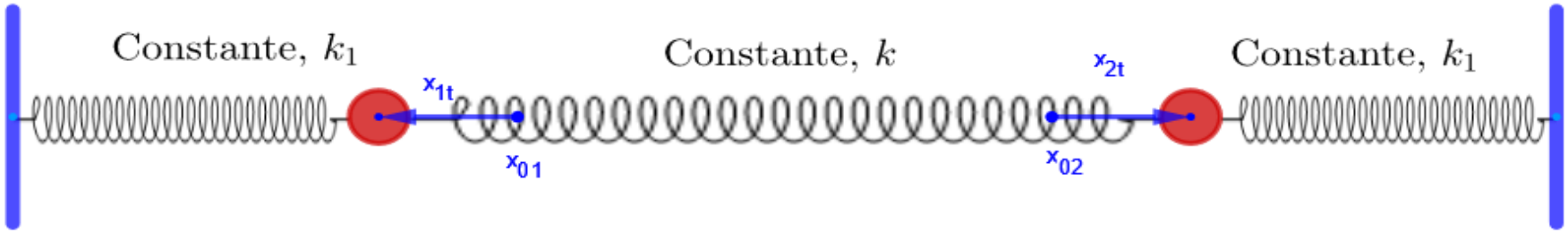


Modo 1

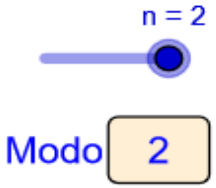
$$x_1(t) = x_2(t) = A\cos(\omega_1 t + \phi_1)$$

$$\omega_1 = \omega_0 = \sqrt{\frac{k_1}{M}}$$





Modo 2

$$x_1(t) = -x_2(t) = A\cos(\omega_2 t + \phi_2)$$

$$\omega_2 = \sqrt{\frac{k_1}{M} + \frac{2k}{M}} = \sqrt{\frac{k_1 + 2k}{M}}$$